

16. ACOG 206: Hormonal Contraception With Coexisting Medical Conditions

Companion reading handout for simultaneous listening.

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Primary source: ACOG Practice Bulletin No. 206, *Use of Hormonal Contraception in Women With Coexisting Medical Conditions*, interpreted through the U.S. Medical Eligibility Criteria framework for contraceptive use.

The difficult part of ACOG 206 is not category 1. Category 1 is the quiet category. No restriction. No meaningful medical barrier. The real clinical work begins when the patient has a disease substrate and the contraceptive method touches that same substrate in some way.

The skill is not to memorize that a condition is category 2, 3, or 4. The skill is to look at a patient, look at a method, and ask: what dangerous pathway is already active in this body, what does this method add, and do those two things move in the same direction?

That is the entire logic of the grey zone.

Category 2 means the risk is real enough that we do not pretend the patient is physiologically normal, but the advantages of the method generally outweigh the theoretical or proven risk.

Category 3 means the risk usually outweighs the advantage. The method is not absolutely forbidden, but it is usually the wrong default. It may still be used if safer or better methods are unavailable, unacceptable, or medically worse for this particular patient.

Category 4 means the condition-method pairing creates an unacceptable health risk. The method should not be used.

Notice the words condition-method pairing. A category is not attached to a disease alone, and it is not attached to a contraceptive alone. It is attached to the collision between them. Migraine with aura is not category 4 for every contraceptive method. It is category 4 for estrogen-containing contraception. Malabsorptive bariatric surgery is not category 3 for every method. It is category 3 for oral contraception, because the oral route is the failing route. Immediate postpartum hemorrhage is not category 4 for the idea of contraception. It is category 4 for IUD insertion into an actively bleeding, diagnostically unresolved uterus.

So the resident should not ask, "What category is this disease?" The correct question is: "What category is this disease with this method, at this moment, in this patient?"

The lecture has one governing model. Classify the problem by mechanism.

There are five method mechanisms that keep returning.

First, estrogen-containing contraception, especially ethinyl estradiol, affects the liver. It increases estrogen-sensitive hepatic proteins, including coagulation factors such as factor VII, factor X, and fibrinogen. It also increases hepatic angiotensinogen, giving the renin-angiotensin-aldosterone system more substrate. That means more potential for angiotensin II and aldosterone physiology: vasoconstriction, sodium retention, oxidative stress, sympathetic facilitation, and less nitric oxide-mediated endothelial reserve. When estrogen becomes dangerous, it is usually because it is being added to a patient who already has thrombosis, arterial disease, severe hypertension, postpartum hypercoagulability, thrombophilia, antiphospholipid antibodies, migraine with aura, or another vascular substrate.

Second, DMPA is not simply "progestin-only." It is a depot, systemic, high-dose, long-acting progestin. It avoids estrogen's hepatic coagulation and RAAS effects, which is why it is often safer than combined hormonal contraception. But it has its own signature: ovulation suppression, hypoestrogenic effects, possible lipid changes, long duration after injection, and early irregular bleeding before amenorrhea is established. DMPA therefore appears in category 3 when arterial disease, severe hypertension, advanced diabetes, or severe thrombocytopenia changes the meaning of those effects.

Third, progestin-only pills, the implant, and LNG-IUDs are often safer in thrombotic disease because they avoid estrogen. But they are not identical. Progestin-only pills are low-dose, oral, and vulnerable to missed doses and enzyme induction. The implant is systemic and very effective, with more pharmacologic reserve. The LNG-IUD is mostly local, with low systemic exposure, but it is still an intrauterine device, so active uterine infection, hemorrhage, or suspected intrauterine trophoblastic disease changes the risk.

Fourth, IUD risk is often not hormonal at all. It is procedural. Is the cavity infected? Is it bleeding? Is there active trophoblastic disease? Is the postpartum uterus involuting? The same device that is excellent in a stable uterus can be inappropriate in an infected, actively hemorrhaging, or diseased uterine cavity.

Fifth, oral methods depend on the gut and on hepatic metabolism. Malabsorptive bariatric surgery threatens absorption. Rifampin, rifabutin, some antiepileptic drugs, and some antiretrovirals threaten steroid exposure by increasing metabolism. These are not toxicity categories. They are failure categories. The method may not hurt the patient directly; it may simply stop being reliable enough.

Keep those five mechanisms in mind: estrogen hepatic vascular biology, DMPA's depot systemic profile, progestin-only method differences, IUD procedural risk, and oral or

systemic steroid exposure being undermined by absorption or enzyme induction.

Now we can reason through the categories.

The Estrogen Question: When Does A Useful Hormone Become The Wrong Signal?

Start with estrogen, because it explains much of category 3 and category 4.

Combined hormonal contraception is a powerful and useful method. In a healthy patient, it suppresses ovulation, stabilizes bleeding, reduces dysmenorrhea, treats some acne and androgen symptoms, reduces ovarian and endometrial cancer risk, and provides highly acceptable contraception for many patients.

But ethinyl estradiol has hepatic effects. The liver increases production of procoagulant proteins. Thrombin-generation potential rises. Sensitivity to natural anticoagulant brakes can shift. Fibrinolytic balance can change. Estrogen also increases angiotensinogen. More angiotensinogen means more substrate for renin, then more potential downstream angiotensin II and aldosterone signaling. Angiotensin II constricts arterioles, promotes aldosterone-mediated sodium retention, facilitates sympathetic tone, increases oxidative stress, and impairs nitric oxide availability. Aldosterone expands volume through sodium retention. Endothelium under oxidative stress becomes less vasodilatory and more inflammatory.

This is why "estrogen raises blood pressure" is too shallow. The deeper statement is that estrogen can increase the substrate feeding RAAS physiology and can shift vascular tone, sodium handling, and endothelial function. In most healthy users, the effect may be small. But if the patient already has hypertension, severe hypertension, vascular disease, postpartum hypercoagulability, thrombophilia, SLE with antiphospholipid antibodies, or migraine with aura, the same biologic nudge lands on a different substrate.

The bedside rule is precise: estrogen is not judged in isolation. Estrogen is judged by the vascular, thrombotic, oncologic, and postpartum substrate it is being added to.

Postpartum Estrogen: The Calendar Is A Clotting Curve

The postpartum estrogen rules are easiest to remember if you stop treating the days as arbitrary. The postpartum calendar is a clotting curve.

Pregnancy prepares the patient not to bleed to death from placental separation. Coagulation capacity rises. Venous capacitance changes. The gravid uterus compresses pelvic veins and the inferior vena cava. Delivery creates tissue injury. The placental bed becomes a broad vascular wound. Cesarean delivery adds surgical injury. Immobility, anemia, pain, transfusion, preeclampsia, infection, and cardiomyopathy may add more risk.

This is Virchow's triad in obstetric form: hypercoagulability, stasis, and tissue injury.

In the first 21 days after birth, combined hormonal contraception is category 4 regardless of breastfeeding status. The reason is not that every postpartum patient will clot. The reason is that early postpartum physiology is already deliberately shifted toward clot formation. Adding estrogen during that peak interval adds hepatic procoagulant drive to a body already in a clot-protective and clot-risk state.

Breastfeeding does not rescue this. A breastfeeding patient in the first 21 days is still postpartum. She still has the same VTE substrate. There may also be a theoretical lactation concern, because estrogen can oppose the hormonal environment that supports early milk production in some patients, but the decisive category 4 mechanism is thrombotic: early postpartum hypercoagulability plus estrogen.

After day 21, the risk is lower, but not gone. That is why breastfeeding patients from day 21 through day 29 remain category 3 for combined hormonal contraception. They are no longer in the category 4 peak, but two systems are still settling. The clotting system has not returned to baseline, and lactation may still be consolidating. Estrogen arrives before either system is fully stable. That is enough to make routine use unfavorable.

ACOG expresses this boundary in overlapping ways: category 4 through postpartum day 21, and category 3 through postpartum day 30 in breastfeeding patients. That overlap teaches the transition. The category is not magic at midnight. It reflects a falling but still clinically relevant postpartum VTE curve, plus theoretical lactation concern.

Then comes days 30 to 42. In a breastfeeding patient with no other VTE risk factors, combined hormonal contraception becomes category 2. Why not category 1? Because postpartum VTE risk has not completely returned to baseline, and breastfeeding remains a physiologic condition where estrogen may theoretically affect milk supply. Why not category 3? Because the first month has passed, lactation is usually more established, and no extra thrombotic driver is present.

Add another VTE risk factor in that same 30 to 42 day breastfeeding window, and the category becomes 3. The added risk factor is not just a label. Each factor has a mechanism. Prior VTE proves susceptibility. Thrombophilia impairs clot regulation. BMI 30 or greater adds venous stasis, inflammation, and metabolic vascular stress. Cesarean delivery adds surgical tissue injury. Preeclampsia adds endothelial dysfunction. Smoking adds oxidative vascular injury and platelet activation. Transfusion often marks severe illness or hemorrhage. Peripartum cardiomyopathy can impair circulation and increase stasis. Estrogen then adds hepatic procoagulant drive.

The rule becomes logical: after 30 days, a breastfeeding patient without other VTE risks is category 2 for CHC; with other VTE risks, category 3.

For the nonbreastfeeding patient, the lactation issue disappears, but the clotting issue remains. Between days 21 and 42, a nonbreastfeeding patient without other VTE risk factors is category 2 for combined hormonal contraception. She may ovulate early, so pregnancy prevention is clinically important. But if she has other VTE risk factors in that same window, the category becomes 3, because postpartum clotting activation plus an added VTE mechanism plus estrogen usually exceeds the benefit of choosing CHC over a safer method.

After 42 days, a nonbreastfeeding patient is category 1 for CHC in the organized reference. A breastfeeding patient more than 42 days after birth is category 2. That difference is not about clotting alone. By that time the VTE risk has fallen substantially, although ACOG notes return to baseline by about 12 weeks. The remaining distinction is breastfeeding. Milk supply is usually established, but estrogen-related lactation concern is not dismissed completely. So the breastfeeding patient beyond 42 days is category 2, not category 1.

Now compare that with progestin-only methods postpartum. Breastfeeding patients less than 30 days after birth are category 2 for progestin-only methods: implant, DMPA, and progestin-only pills. Why category 2? Because the first month postpartum is biologically active. Lactogenesis, uterine healing, and VTE risk are not trivial. Why not category 3? Because progestin-only methods avoid estrogen's VTE mechanism and available breastfeeding and infant-growth data are reassuring, including data suggesting no major harm to initiation, continuation, growth, or development. The theoretical concern is progesterone physiology around lactogenesis. The clinical evidence does not justify routine avoidance.

The postpartum lesson is therefore compact.

Early postpartum plus estrogen is the problem. First 21 days is category 4. Breastfeeding day 21 to day 29 is category 3. Day 30 to 42 depends on added VTE risk. Beyond 42 days, breastfeeding keeps CHC at category 2, while nonbreastfeeding becomes category 1. Progestin-only methods avoid the estrogen clotting signal and are much more favorable.

Surgery: Immobilization Is The Missing Limb Of Virchow's Triad

Major surgery with anticipated prolonged immobilization is category 4 for combined hormonal contraception.

The mechanism is direct. Surgery creates tissue injury, tissue factor exposure, endothelial disruption, inflammatory cytokine signaling, platelet activation, and postoperative coagulation activation. Immobilization adds venous stasis. The calf muscle pump is inactive; venous flow slows; activated clotting factors are less diluted and cleared; platelets, leukocytes, coagulation proteins, and endothelium interact longer in low-flow veins.

Estrogen adds hepatic procoagulant effects. It does not need to cause stasis itself. It makes the coagulation side more active while surgery and immobilization create the perfect environment for clot formation.

This is category 4 because all three limbs converge: tissue injury, stasis, and hypercoagulability.

Now remove prolonged immobilization. Major surgery without expected prolonged postoperative immobilization is category 2 for CHC. The patient still has surgical tissue injury and coagulation activation, so it is not category 1. Estrogen still contributes procoagulant biology. But the prolonged stasis limb is absent, and ACOG states there is no reason to stop combined hormonal contraception when the patient is expected to be ambulatory.

The detail about timing matters. ACOG notes that clotting factors do not normalize unless CHC is discontinued four to six weeks before major surgery. Stopping the pill the day before surgery is not a meaningful biologic reversal. So the decision is not theatrical medication holding. The decision is whether the planned surgery includes prolonged immobilization. If yes, category 4. If no, category 2.

Thrombophilia And Prior VTE: Is The Anticoagulant Brake Broken, Or Was The Old Trigger Temporary?

Known familial thrombophilias are category 4 for combined hormonal contraception. These include conditions such as factor V Leiden, prothrombin G20210A mutation, protein C deficiency, protein S deficiency, and antithrombin deficiency.

The mechanisms differ, but they share one endpoint: thrombin is generated too easily or controlled too poorly.

Factor V Leiden makes factor V resistant to activated protein C. Normally, activated protein C, with protein S as cofactor, turns off factors Va and VIIIa. If factor V resists that brake, the prothrombinase complex remains active longer and generates more thrombin.

Prothrombin G20210A increases prothrombin levels. More substrate is available for conversion to thrombin. Protein C and protein S deficiencies weaken the pathway that limits factors Va and VIIIa. Antithrombin deficiency weakens inhibition of thrombin and factor Xa. In each case, the system loses a natural brake.

Estrogen then increases procoagulant hepatic proteins and shifts thrombin generation upward. ACOG cites the clinical magnitude: heterozygous factor V Leiden carries about a sevenfold higher baseline VTE risk than noncarriers, and more than 15 to 30 times higher risk in heterozygous factor V Leiden carriers using oral contraceptives than in noncarriers using nonhormonal contraception.

That is not a mild interaction. That is why known familial thrombophilia plus CHC is category 4.

But known thrombogenic mutations with progestin-only methods and LNG-IUDs are category 2. The baseline substrate remains real; therefore this is not category 1. But the selected methods avoid estrogen's hepatic coagulation trigger. LNG-IUD is mostly local. Progestin-only methods vary in systemic exposure, but they do not add the same estrogen-driven clotting signal. Pregnancy itself is hypercoagulable. So category 2 is a comparative-risk judgment: baseline thrombophilia is real, but these methods are usually safer than pregnancy and safer than estrogen-containing contraception.

Now consider a remote single VTE from a nonrecurring risk factor. ACOG gives combined hormonal contraception category 3 in the narrow scenario of a single VTE episode years earlier, associated with a nonrecurring risk factor, if no other method is available or desired.

This is a classic grey-zone case. A remote provoked clot is not the same as known thrombophilia or recurrent unprovoked VTE. The old clot may have resulted from a temporary convergence: trauma, immobilization, surgery, or another resolved trigger. Once the trigger is gone, recurrence risk may be lower.

But the old clot is not meaningless. It proves that this patient's venous system has crossed into pathologic thrombosis under stress. There may be residual venous valve damage, altered flow, or unrecognized predisposition. Estrogen would reintroduce a prothrombotic exposure.

So why category 3 rather than 4? Because the provoking factor was remote and nonrecurring. Why not category 2? Because estrogen still acts on a patient with demonstrated VTE susceptibility. The correct teaching line is: a remote provoked VTE lowers recurrence concern compared with persistent thrombophilia, but it does not erase the meaning of the event.

SLE: The Antibody Status Changes The Disease

SLE is not a single contraceptive category. The category depends on the vascular and immunologic phenotype.

SLE with positive antiphospholipid antibodies is category 4 for combined hormonal contraception. The mechanism is immunothrombosis plus estrogen.

Antiphospholipid antibodies often target phospholipid-binding proteins such as beta-2 glycoprotein I. These antibody-protein complexes activate endothelial cells, monocytes, platelets, neutrophils, complement, and tissue factor pathways. Endothelium becomes more adhesive and procoagulant. Monocytes express tissue factor. Platelets become more reactive. Complement amplifies inflammation and vascular injury. Natural anticoagulant surface mechanisms may be impaired. Nitric oxide signaling may be reduced.

This is not generic clot risk. It is immune-driven coagulation activation. Add estrogen-driven hepatic procoagulant effects, and the result is category 4.

Now change the method. SLE with antiphospholipid antibodies is category 3 for progestin-only methods, including LNG-IUDs. This is one of the hardest category 3 concepts. The method is not the major thrombosis amplifier; the patient is. Progestin-only methods do not have estrogen's hepatic coagulation effect, and LNG-IUD exposure is mostly local. That is why the category is not 4. But antiphospholipid-positive SLE is so thrombotic at baseline that routine hormonal use should not be casual. Category 3 says: default away, but preserve individualized use when alternatives are unacceptable.

Now remove antiphospholipid antibodies. SLE without antiphospholipid antibodies is category 2 for progestin-only methods. The patient still has systemic inflammatory disease, possible nephritis, steroid exposure, hypertension, and vascular complexity, so not category 1. But the decisive antiphospholipid immunothrombotic amplifier is absent, and progestin-only methods avoid estrogen. Category 2 is the bounded-risk conclusion.

SLE without antiphospholipid antibodies and without other cardiovascular disease risk factors is category 2 for combined hormonal contraception. This is a carefully narrowed subgroup. The exclusions do the work: no antiphospholipid antibodies, no smoking, no hypertension, no diabetes, no hypercholesterolemia, no older-age cardiovascular risk. ACOG also notes that CHC does not appear to worsen SLE disease activity in women with inactive or stable active disease. Estrogen is not ignored; that is why the category is 2 rather than 1. But the thrombotic amplifiers that would make estrogen usually inappropriate are absent.

SLE on immunosuppressant medications with LNG-IUD is category 2. The theoretical concern is infection: an intrauterine foreign body in an immunosuppressed patient. But ACOG cites safety data without increased infection, contraceptive failure, or other adverse events in immunosuppressed women using LNG-IUDs. The method is local and estrogen-free. So category 2 means: use is generally favorable, but insertion must still respect ordinary infection screening and active pelvic infection contraindications.

Finally, SLE with severe thrombocytopenia under 50,000 platelets per microliter is category 3 for DMPA initiation. This is not a thrombosis rule; it is a bleeding rule. Platelets form the primary hemostatic plug and provide a phospholipid surface for coagulation enzyme complexes. Below 50,000, bleeding risk becomes clinically important, especially with mucosal bleeding or uterine bleeding.

DMPA can ultimately produce amenorrhea, which could help bleeding. But initiation is the unstable phase. Early DMPA use can cause irregular, prolonged, or unscheduled bleeding as the endometrium thins, superficial vessels become fragile, and repair is unstable before amenorrhea is established. In severe thrombocytopenia, the uterus has less platelet capacity to seal those bleeding points. That is why initiation is category 3. The timing is the mechanism.

Migraine: Aura Is Not A Headache Detail

Migraine with aura is category 4 for estrogen-containing contraception.

The reason is not headache severity. Aura identifies a neurovascular phenotype associated with increased ischemic stroke risk. Aura is thought to involve cortical spreading depolarization: a wave of neuronal and glial depolarization followed by changes in cerebral blood flow and excitability. Clinically, it produces transient focal neurologic symptoms, often visual, sensory, speech-related, or motor.

The contraceptive issue is that migraine with aura marks a circulation with stroke vulnerability. Mechanistic links include endothelial dysfunction, abnormal vasoreactivity, platelet activation, inflammatory vascular signaling, and reduced vascular reserve. Estrogen adds arterial thrombotic potential. ACOG cites ischemic stroke risk being highest in women with migraine with aura using combined hormonal contraception, with an adjusted odds ratio of 6.1 compared with women with neither migraine nor CHC use.

So migraine with aura plus estrogen is category 4. It is not about the pain. It is about focal neurologic aura as a stroke-risk phenotype.

Migraine without aura and no other stroke risk factors is category 2 for combined hormonal contraception. Migraine without aura lacks the focal neurologic aura phenotype most strongly linked to ischemic stroke. ACOG notes it accounts for about 75 percent of migraine cases. Estrogen still has arterial thrombotic implications, so the category is not 1. But without aura and without other stroke risk factors, the stroke substrate is not strong enough to make CHC usually inappropriate. In menstrual migraine, extended hormonal contraception may even reduce estrogen-withdrawal migraine by stabilizing hormone fluctuations.

The resident should hear the contrast clearly: aura changes the category because aura changes the vascular meaning of the migraine.

Hypertension: Estrogen Presses On The Same Axis That Is Already Abnormal

Hypertension is where vague teaching is especially dangerous. Do not say only that estrogen can slightly raise blood pressure. Explain why.

In hypertension, the arterial system is exposed to elevated wall stress. Endothelium shifts toward reduced nitric oxide, oxidative stress, inflammation, and increased permeability to atherogenic lipoproteins. Smooth muscle hypertrophies. Arteries stiffen. Left ventricular afterload rises. Cerebral, renal, and coronary vessels become more vulnerable.

Ethinyl estradiol can increase hepatic angiotensinogen synthesis. Angiotensinogen is the substrate for renin. More substrate can support more angiotensin I, then angiotensin II. Angiotensin II constricts arterioles, promotes aldosterone secretion, increases sodium retention, facilitates sympathetic tone, increases oxidative stress, and worsens endothelial function. Estrogen also shifts coagulation toward thrombosis.

For BP 140 to 159 systolic or 90 to 99 diastolic, combined hormonal contraception is category 3. This is established hypertension. It is not category 4 because the pressure is not at the severe threshold and vascular disease is not specified. But it is not category 2 because estrogen pushes on the exact vascular and RAAS pathways already abnormal. ACOG notes contemporary low-dose combined OCs can raise BP by about 8 mm Hg systolic and 6 mm Hg diastolic compared with copper IUD users, and hypertensive CHC users have increased MI risk. In a patient already hypertensive, even a modest rise can matter.

Adequately controlled hypertension on medication is also category 3 for combined hormonal contraception. Controlled numbers do not mean normal vascular biology. Medication lowers measured pressure, but it may not erase arterial stiffness, endothelial dysfunction, renal sodium sensitivity, RAAS tendency, obesity, family predisposition, or metabolic risk. Estrogen can challenge that controlled equilibrium through angiotensinogen and procoagulant effects. ACOG leaves room for clinical judgment in carefully selected young, otherwise healthy, nonsmoking patients without end-organ disease, with close BP monitoring. But the default category remains 3.

Severe hypertension, systolic 160 or greater or diastolic 100 or greater, or vascular disease, is category 4 for combined hormonal contraception. Here the arterial wall stress is high enough, or vascular injury established enough, that adding estrogen becomes unacceptable. Estrogen may worsen RAAS-mediated pressure biology and add thrombotic tendency to arteries already exposed to high mechanical stress or existing disease. ACOG cites a 12-fold higher relative risk of acute myocardial infarction among hypertensive CHC users compared with nonusers.

Now compare DMPA. Severe hypertension is category 3 for DMPA, not 4. Why? DMPA does not contain estrogen and does not add the same RAAS and hepatic coagulation signature. But DMPA can produce hypoestrogenic effects and unfavorable lipoprotein changes. In a severe hypertensive patient, lipids matter because high pressure injures endothelium and promotes LDL entry into the arterial wall. LDL can oxidize, be taken up by macrophages, and become foam cells. HDL supports reverse cholesterol transport and endothelial protection. If DMPA worsens the lipid environment, it may add arterial risk to a high-pressure vascular bed.

So CHC is category 4 in severe hypertension because it can worsen the pressure and thrombosis axis directly. DMPA is category 3 because its concern is more indirect: lipid and hypoestrogenic vascular effects in a high-risk arterial substrate.

Diabetes: Uncomplicated Disease Is Not The Same As Vascular Disease

Diabetes teaches another threshold. Uncomplicated insulin-dependent or noninsulin-dependent diabetes is category 2 for hormonal contraception. Diabetes can injure vessels through hyperglycemia, advanced glycation end products, oxidative stress, endothelial dysfunction, platelet activation, and dyslipidemia. That is why it is not

category 1. But ACOG's uncomplicated category means no duration beyond 20 years and no retinopathy, nephropathy, neuropathy, or other vascular disease. Available data are reassuring on carbohydrate metabolism. Pregnancy in diabetes can be medically consequential. So the advantages of effective contraception generally outweigh the risk.

Now move to diabetes for more than 20 years or diabetes with microvascular disease. DMPA becomes category 3. Retinopathy, nephropathy, and neuropathy are not isolated organ trivia. They prove systemic vascular injury. Retinopathy reflects retinal microvascular damage. Nephropathy reflects glomerular endothelial and basement membrane injury, often with albuminuria and increased cardiovascular risk. Neuropathy reflects metabolic and microvascular injury to nerves. Duration over 20 years means prolonged exposure to glycation, oxidative stress, endothelial dysfunction, and inflammatory vascular signaling.

DMPA may worsen lipoprotein profiles and may reduce endogenous estrogenic vascular support. In advanced diabetes, that matters. The arterial and microvascular substrate is already damaged. Category 3 says DMPA is usually not worth adding when other progestin-only methods, implant, LNG-IUD, copper IUD, or other options can avoid this particular DMPA lipid and hypoestrogenic concern.

Combined hormonal contraception in diabetes for more than 20 years or with microvascular disease is listed by ACOG as category 3 or 4 depending on severity. The reason is the same vascular substrate, but estrogen adds procoagulant and RAAS-related vascular stress. The exact category depends on the severity and associated vascular disease.

Age, Obesity, And Rheumatoid Arthritis: Bounded Risk Does Not Mean No Risk

A healthy nonsmoking patient age 50 to 55 without cardiovascular risk factors is category 2 for continuing combined hormonal contraception. Age increases baseline risk of VTE, hypertension, and arterial events. Fertility declines but does not disappear reliably until menopause. CHC can provide perimenopausal benefits: cycle control, reduced abnormal bleeding, vasomotor symptom improvement, bone mineral density support, and reduced ovarian and endometrial cancer risk. So age alone is not a command to stop. It is a command to reassess. Category 2 means continuation can be reasonable when the patient is healthy, nonsmoking, and free of cardiovascular risk factors.

Age older than 45 without other cardiovascular disease risk factors is category 2 for DMPA. The issue is not estrogen thrombosis. DMPA is estrogen-free. The concern is that DMPA can lower endogenous estradiol, and the patient is approaching menopause, when bone mineral density and vascular risk become more important. ACOG notes uncertainty about whether bone mineral density in long-term DMPA users returns to baseline before menopause. The risk is real enough to avoid category

1, but without added cardiovascular risk factors it does not usually outweigh DMPA's benefit.

Obesity is category 2 for combined hormonal contraception. Obesity increases VTE risk through venous stasis, chronic low-grade inflammation, adipokines, endothelial dysfunction, and associated metabolic risk. It may alter steroid pharmacokinetics; ACOG notes women with obesity may require twice as long to reach steady-state therapeutic steroid levels after starting the pill or after the hormone-free interval. CHC adds estrogen-related procoagulant effects. Why not category 3? Because obesity alone does not usually make the absolute CHC risk exceed the benefit of effective contraception, and ACOG emphasizes that VTE risk with CHC in women with obesity remains less than VTE risk during pregnancy and the puerperium in women with obesity. Category 2 is a comparative-risk conclusion.

Rheumatoid arthritis, in ACOG's multiple-condition example, is category 2 for combined hormonal contraception. Rheumatoid arthritis is systemic inflammation, and depending on severity and treatment it may involve reduced mobility, corticosteroids, cardiovascular risk, or immunosuppressive therapy. That is why it is not category 1. But isolated rheumatoid arthritis does not by itself create the specific high-risk mechanisms that dominate CHC: aura-related stroke phenotype, severe hypertension, antiphospholipid antibodies, known thrombophilia, or active VTE. In ACOG's example, the final decision is controlled by the coexisting migraine with aura, which is category 4. The categories do not average. The highest-risk condition controls.

DMPA And Progestin-Only Methods: Do Not Treat Them As One Bucket

One of the most important resident-level skills in ACOG 206 is to stop saying "progestin-only" as if it were a single biologic exposure.

Progestin-only pills are low-dose, oral, and timing-sensitive. The implant is systemic, continuous, and very effective. DMPA is systemic, depot, high-dose, ovulation-suppressing, and long-acting. The LNG-IUD is primarily local, with low systemic levonorgestrel exposure, and it is also a device in the uterine cavity.

Those differences explain many category shifts.

Prior VTE, current VTE risk, prior MI, or prior stroke is category 2 for initiation of progestin-only pills, implant, or LNG-IUD. The substrate is serious: venous thrombosis, arterial ischemia, or active risk. It is not category 1. But the methods avoid estrogen's hepatic coagulation and RAAS effects. LNG-IUD is mostly local. Pregnancy itself may be more dangerous than these methods. Therefore category 2.

VTE risk with DMPA is also category 2, but with a different level of caution. DMPA avoids estrogen, which is the central reason it is not category 3 in many VTE-risk

patients. But ACOG notes DMPA data are less clean than for other progestin-only methods. A meta-analysis based on two studies found a twofold increased VTE risk with progestin-only injectables, although residual confounding limits certainty. DMPA is systemic, long-acting, and cannot be rapidly withdrawn. Category 2 means use is generally reasonable, but DMPA is not as mechanistically clean as LNG-IUD or implant in VTE-risk reasoning.

New DVT, ischemic cardiac event, or stroke developing during use of a progestin-only pill, implant, or LNG-IUD is category 3 for continuation. The key word is new. The method may have been appropriate before the event. But the event changes the risk state. Acute thrombosis or ischemia may involve active clot propagation, endothelial injury, platelet activation, immobility, new anticoagulation, diagnostic uncertainty, occult thrombophilia, malignancy, antiphospholipid syndrome, or vascular disease.

This category does not say the method caused the event. It says the event forces a stop-and-recalculate moment. Stabilize the patient, determine the mechanism, reassess pregnancy risk, and decide whether a nonhormonal or lower-exposure method is better. It is not category 4 because these methods do not contain estrogen and are not strong thrombotic amplifiers. It is category 3 because continuing automatically would treat the old risk state as if it still existed.

Personal history of ischemic heart disease or stroke is category 3 for DMPA. This is arterial disease. The patient has already had a failure of coronary or cerebral perfusion. DMPA does not add estrogen, so it is not category 4 in this ACOG statement. But DMPA may lower endogenous estrogenic support and worsen lipid profiles. Endogenous estrogen supports nitric oxide production, endothelial function, favorable lipid effects, and vascular repair. If HDL falls, reverse cholesterol transport and anti-inflammatory endothelial effects may weaken. If LDL or total cholesterol rises, more atherogenic substrate is available for arterial wall deposition, oxidation, macrophage uptake, foam-cell formation, and plaque progression. In a patient with prior MI or stroke, this matters. Category 3 reflects arterial substrate plus DMPA's lipid and hypoestrogenic concern.

Advanced diabetes and severe hypertension with DMPA follow the same logic: DMPA is not estrogen, but its lipid and hypoestrogenic profile can matter when the arterial system is already injured.

Anticoagulant therapy is category 2 for all progestin-only methods. This is a paradoxical substrate: the patient is thrombotic enough to need anticoagulation, but anticoagulation increases bleeding risk. Warfarin is teratogenic, so preventing pregnancy matters. Anticoagulation can cause heavy menstrual bleeding and, rarely, hemoperitoneum after ovarian cyst rupture.

Progestin-only methods avoid estrogen's VTE effect, and some directly help the gynecologic bleeding problem. The 52-mg LNG-IUD reduces menstrual blood loss substantially. DMPA suppresses ovulation and may reduce hemorrhagic ovarian cyst risk. Progestin-only pills and implants may cause irregular bleeding, so they require

counseling, but they do not add estrogen-driven VTE risk. Category 2 means the patient is complex, not that the method is usually wrong.

Therapeutic anticoagulation with combined hormonal contraception is category 3 when alternatives are not acceptable or are contraindicated. This is a true risk-benefit category 3. CHC can reduce heavy menstrual bleeding and suppress ovulation, which may reduce anticoagulation-associated bleeding and hemorrhagic cysts. But CHC also adds estrogen-related thrombotic biology to a patient who needs anticoagulation because of thrombotic risk. Therapeutic anticoagulation may lower recurrent thrombosis risk, but only while it is therapeutic and uninterrupted. Warfarin time in range, adherence, interactions, renal function, procedures, and interruptions all matter. So CHC may be considered case by case, but the default is category 3.

The resident should separate two questions. If the goal is contraception and bleeding control in an anticoagulated patient, progestin-only methods, especially LNG-IUD or DMPA, often make physiologic sense. If considering CHC, the question becomes whether the bleeding benefit justifies adding estrogen to a thrombotic patient whose protection depends on reliable anticoagulation.

Breast Cancer And Hormonal Signaling: The Risk Is Oncologic, Not Immediate Toxicity

Current or recent breast cancer is category 4 for combination estrogen-progestin contraception, progestin-only contraception, and LNG-IUD.

The mechanism is not clotting. It is hormone signaling in malignancy. Many breast cancers are estrogen receptor positive or progesterone receptor positive. Estrogen receptor signaling can activate transcriptional programs for proliferation, cell-cycle progression, survival, and tumor growth. Progesterone receptor biology interacts with breast epithelial proliferation and tumor behavior. Treatments such as tamoxifen, aromatase inhibitors, and ovarian suppression exist because hormone signaling can sustain tumor growth or recurrence.

Combined hormonal contraception adds estrogen and progestin. Progestin-only methods remove estrogen but still add progestin signaling. LNG-IUD is mostly local, but systemic levonorgestrel exposure is not zero, and in current or recent breast cancer the question is not merely serum concentration. It is whether exogenous hormonal signaling could stimulate residual hormone-sensitive malignant cells or micrometastatic disease during a clinically meaningful recurrence-risk interval.

That is category 4.

No evidence of breast cancer disease for five years or more after treatment is category 3 for hormonal contraception, including combination estrogen-progestin contraception, progestin-only contraception, and LNG-IUD. The disease-free interval lowers risk compared with current or recent disease, but it does not make the biologic

risk zero. Dormant micrometastatic cells may persist below detection. Hormone receptor-positive breast cancer can recur late. Exogenous hormonal signaling remains theoretically concerning.

So why not category 4? Because the time without evidence of disease lowers the immediate oncologic concern. Why not category 2? Because nonhormonal options, especially copper IUD, avoid the hormone-signaling question. Category 3 says use nonhormonal contraception if possible; consider hormonal methods only after individualized oncologic and contraceptive reasoning.

Treated breast cancer with copper IUD is category 1 in the organized reference because copper IUD provides contraception without estrogen or progestin signaling. That contrast teaches the mechanism. The problem is not contraception. The problem is hormonal stimulation in a hormone-sensitive cancer context.

Gynecologic cancers are different. Cervical, endometrial, or ovarian cancer while awaiting surgery, or when treatment does not cause sterility, may have combined hormonal or progestin-only contraception including continuation of a previously placed LNG-IUD as category 1 or 2. The reason is that contraception may be needed temporarily, treatment often causes sterility, and the hormone concern depends on cancer type, treatment plan, and whether a method is initiated or continued. Do not transfer the breast cancer rule mechanically to every cancer.

IUDs In An Unsafe Uterus: The Device Is The Problem, Not The Contraceptive Goal

IUD categories are sometimes about the uterine cavity rather than hormones.

Immediate postpartum LNG-IUD placement in a breastfeeding patient immediately after placental delivery is category 2, unless contraindications exist. The uterus is large, recently emptied, and beginning involution. The cervix is open. The placental bed is a raw vascular surface. Expulsion risk is higher than interval insertion. That is why it is not category 1. But it avoids estrogen, provides immediate highly effective contraception, and prevents the common failure point of missed postpartum follow-up. That is why it is not category 3.

Breastfeeding or nonbreastfeeding patients up to four weeks postpartum are category 2 for LNG-IUD, assuming no contraindications. The postpartum uterus is still involuting, lochia and endometrial recovery are ongoing, and expulsion risk remains relevant. But if infection and hemorrhage are absent, the IUD's high efficacy and estrogen avoidance generally outweigh those mechanical risks.

At four or more weeks after birth, breastfeeding or nonbreastfeeding patients are category 1 for LNG-IUD in the organized reference. By then the postpartum uterine mechanical issues have receded enough for no restriction in the referenced classification.

Now contrast the category 4 states.

Immediate postpartum uterine infection is category 4 for IUD insertion. Peripartum chorioamnionitis, postpartum endometritis, or puerperal sepsis means infection and inflammation are already present in the reproductive tract. The postpartum uterine cavity contains lochia and healing decidua. The placental site is a raw vascular wound. The cervix has recently been open. Instrumentation can move organisms, disturb infected tissue, and introduce a foreign body into an infected space. A foreign body can support bacterial adherence and biofilm formation. The risk is worsening uterine infection, ascending infection, pelvic infection, or sepsis. Category 4 here is time-limited: do not insert into an infected postpartum uterus before resolution and evaluation.

Immediate postpartum ongoing hemorrhage is category 4 for IUD insertion. Ongoing hemorrhage means the uterus and patient are unstable until proven otherwise. The differential includes atony, retained tissue, genital tract laceration, uterine rupture, uterine inversion, and coagulopathy. Each has a different treatment pathway. IUD insertion is nonessential and can interfere with exploration, ultrasound assessment, balloon tamponade, curettage, or hemorrhage control. It may be expelled from an atonic or actively contracting uterus, may worsen bleeding, or may obscure the source. Category 4 means do not insert a device into an actively bleeding, unresolved postpartum uterus.

Gestational trophoblastic disease with persistently elevated hCG or malignant disease when intrauterine disease is evident or suspected is category 4 for new IUD placement. GTD is abnormal proliferation of trophoblastic tissue. hCG is the surveillance marker. Persistently elevated hCG means active trophoblastic tissue may remain or malignant disease may be present. If intrauterine disease is evident or suspected, the cavity may contain vascular, friable abnormal tissue. IUD insertion passes an instrument through the cervix into that cavity and places a foreign body there. It can provoke bleeding, perforate abnormal tissue, introduce infection, or complicate interpretation of bleeding.

Contraception is important in GTD because a new pregnancy confuses hCG surveillance. The category 4 issue is not contraception. It is new IUD placement when active intrauterine disease is evident or suspected.

Hormonal methods after suction curettage for GTD, when hCG is falling or undetectable, or when hCG is elevated but intrauterine disease is not evident or suspected, are category 1 or 2. That contrast is critical. Hormonal contraception does not instrument the uterine cavity. IUD placement does.

Drug Interactions And Absorption: Sometimes The Harm Is Failure

Some category 3 and category 2 rows are not about toxicity. They are about whether the contraceptive can still work.

Malabsorptive bariatric surgery, such as Roux-en-Y gastric bypass or biliopancreatic diversion, is category 3 for oral contraception, whether combined hormonal or progestin-only. The method does not become toxic. It becomes unreliable.

Roux-en-Y creates a small gastric pouch and bypasses the duodenum and part of the proximal jejunum. Biliopancreatic diversion produces more substantial malabsorption by separating food flow from biliopancreatic enzymes for a longer segment. Oral contraceptives require dissolution, absorption, first-pass handling, and systemic hormone exposure. Vomiting, diarrhea, altered pH, reduced absorptive surface, bypassed proximal bowel, shortened contact time, and altered enterohepatic cycling can make hormone levels less predictable.

If estrogen or progestin exposure falls below the threshold needed for ovulation suppression, cervical mucus effects, and endometrial effects, pregnancy can occur. After bariatric surgery, pregnancy timing matters because early postoperative pregnancy may occur during rapid weight loss, nutritional instability, and micronutrient deficiency risk. Nonoral methods bypass the damaged absorption pathway and are category 1 in the organized reference.

Restrictive bariatric surgery, such as vertical banded gastroplasty, laparoscopic adjustable gastric band, or sleeve gastrectomy, is category 1 for oral contraception in the reference because it does not create the same bypassed absorptive anatomy.

Rifampin or rifabutin use is category 3 for progestin-only pills. Rifampin especially is a strong inducer of hepatic and intestinal drug-metabolizing pathways, including CYP3A-related metabolism and transporter systems. Progestin-only pills are low-dose oral methods. Traditional norethindrone pills depend heavily on cervical mucus thickening and variably suppress ovulation. Drospirenone progestin-only pills suppress ovulation more consistently but still depend on systemic progestin exposure. If rifamycins lower progestin levels, cervical mucus may become less hostile to sperm, ovulation suppression may fail, and endometrial effects may weaken.

This is category 3 because the pill does not have enough pharmacologic reserve against a strong inducer.

Rifampin use with the etonogestrel implant is category 2. Why not category 1? Rifampin is a strong inducer and can plausibly lower etonogestrel exposure. Why not category 3? The implant starts from extremely high contraceptive efficacy, releases hormone continuously, and avoids daily adherence failures. It has more pharmacologic reserve than a progestin-only pill. DMPA and LNG-IUD are category 1 with rifampin or rifabutin in the organized reference because ACOG states there is no evidence of decreased efficacy, and these methods should be encouraged for long-term rifamycin use.

Enzyme-inducing antiepileptic medications with combined hormonal contraception are category 3. ACOG names drugs such as carbamazepine, felbamate, oxcarbazepine, phenobarbital, phenytoin, primidone, and rufinamide. These increase metabolism of contraceptive steroids. Combined hormonal contraception requires adequate estrogen and progestin exposure to suppress the hypothalamic-pituitary-ovarian axis, prevent the LH surge, suppress ovulation, thicken cervical mucus, and alter endometrial receptivity. If steroid levels fall, breakthrough bleeding, follicular development, breakthrough ovulation, and pregnancy may occur.

In epilepsy, failure is high stakes. Pregnancy may expose the fetus to teratogenic antiepileptic drugs, and seizure control can change during pregnancy. ACOG notes fetal antiepileptic drug exposure is associated with a twofold to threefold increased risk of major congenital malformations compared with the general population, with higher rates reported with valproate or polytherapy.

ACOG mentions mitigation if CHCs are used despite the interaction: 30 to 35 micrograms rather than lower ethinyl estradiol doses, longer half-life progestins, and hormone-free intervals shorter than seven days. These strategies make mechanistic sense. More estrogen gives more substrate before induction lowers levels. Longer half-life progestins maintain concentrations longer. Shorter hormone-free intervals reduce the window for follicular recruitment. But mitigation is not the same as removing the interaction. The default remains category 3.

Enzyme-inducing antiepileptic drugs with progestin-only pills are also category 3. In fact, progestin-only pills are especially vulnerable because serum norethindrone levels during progestin-only pill use are lower than during combined OC use. Lower progestin may mean loss of cervical mucus effect and ovulation suppression. DMPA and IUDs are preferred and category 1 in the reference because DMPA delivers high sustained progestin levels and IUDs avoid systemic steroid dependence.

Enzyme-inducing antiepileptic medications with the etonogestrel implant are category 2. Case reports of pregnancy and pharmacokinetic evidence with carbamazepine show the interaction is real, so not category 1. But the implant's continuous release, very high baseline efficacy, and lack of daily adherence dependence mean the interaction is generally outweighed by method effectiveness, so not category 3. DMPA or IUD may still be preferable when maximal certainty is required.

St. John's wort with CHC, progestin-only pills, and etonogestrel implant is category 2. St. John's wort can induce steroid metabolism and ACOG cites breakthrough bleeding and increased likelihood of ovulation in combined OC users. The reason it is category 2 rather than 3 is that its induction signal is less dominant and less predictable than rifampin or major enzyme-inducing antiepileptics. But it is not category 1 because an over-the-counter supplement can weaken contraceptive steroid exposure. Residents must ask about it because patients may not volunteer it as a medication.

Rifampin, enzyme-inducing antiepileptics, and interacting antiretrovirals with combined hormonal contraception or progestin-only pills are category 3 when they

interfere with steroid efficacy. Antiretroviral therapy excluding fosamprenavir is category 1 or 2 for all hormonal methods in the organized reference, because not every regimen produces a clinically dominant interaction. The method and regimen must be named. Do not say "HIV medications interact" as a blanket rule.

The memory line is this: if the problem is absorption or enzyme induction, ask which route and how much pharmacologic reserve the method has. Oral low-dose methods are fragile. Depot DMPA and IUDs are resilient. The implant is in between: strong, but not completely immune.

The Category 2 Mindset: Bounded Risk With A Better Alternative To Pregnancy

Category 2 is not "no issue exists." It is "the advantages generally outweigh the risk, but the risk has a name."

This matters because category 2 is where sloppy teaching can make residents either too casual or too fearful.

Consider VTE risk with progestin-only methods, thrombogenic mutations with progestin-only methods and LNG-IUDs, SLE without antiphospholipid antibodies, obesity with CHC, migraine without aura with CHC, uncomplicated diabetes with hormonal contraception, age over 45 with DMPA, and major surgery without prolonged immobilization with CHC. In each case there is a real substrate. That substrate prevents category 1. But the dangerous mechanism is either absent, bounded, local, weaker, or outweighed by pregnancy prevention.

The category 2 question should be asked in four steps.

First, what is the substrate? Prior clot, obesity, postpartum state, diabetes, migraine, age, surgery, inflammatory disease, anticoagulation, enzyme induction, or lactation.

Second, what does the method add? Estrogen hepatic effect, DMPA lipid/hypoestrogenic effect, oral exposure vulnerability, local uterine device effect, systemic progestin exposure, or bleeding pattern.

Third, why is this not category 1? There must be a real physiologic reason.

Fourth, why is this not category 3? The risk must remain bounded: wrong mechanism absent, magnitude lower, route safer, exposure local, method buffered, or pregnancy risk greater.

Take obesity with CHC. The substrate is VTE risk from stasis, inflammation, endothelial dysfunction, and metabolic risk. CHC adds estrogen. Not category 1. But obesity alone does not usually make absolute risk high enough to outweigh the benefit of effective contraception, especially compared with pregnancy and puerperium. Category 2.

Take migraine without aura with CHC. The substrate is migraine, but not the aura phenotype. Estrogen adds arterial thrombotic implications. Not category 1. But without aura and without other stroke risk factors, the strong stroke signal that makes migraine with aura category 4 is absent. Category 2.

Take major surgery without immobilization and CHC. Surgery adds tissue injury and inflammation. Estrogen adds procoagulant hepatic effects. Not category 1. But expected ambulation removes the prolonged venous stasis limb. Category 2.

Take anticoagulation with progestin-only methods. The substrate is paradoxical: thrombosis indication plus medication-induced bleeding risk plus teratogenic risk if on warfarin. Progestin-only methods avoid estrogen, and LNG-IUD or DMPA may help bleeding or ovarian cyst risk. Not category 1 because bleeding and thrombosis are both real. Not category 3 because the method biology often helps more than it harms. Category 2.

This is how category 2 should feel: not trivial, but generally usable.

The Category 3 Mindset: Usually Avoid, But Do Not Turn It Into Category 4

Category 3 is the hardest category because it demands disciplined nuance. It is not a contraindication. It is also not a casual permission category. It means the risks usually outweigh the advantages.

There are three types of category 3 in ACOG 206.

The first is toxicity or physiologic harm category 3. Examples include CHC with moderate hypertension, DMPA with prior ischemic heart disease or stroke, DMPA with severe hypertension, DMPA with advanced diabetes, progestin-only methods in antiphospholipid-positive SLE, DMPA initiation in severe thrombocytopenia, and hormonal contraception after five disease-free years from breast cancer.

The second is cumulative-risk category 3. Examples include postpartum CHC after the category 4 window but before risk has normalized, remote provoked VTE with CHC, and CHC in therapeutic anticoagulation when the bleeding benefit competes with thrombotic concern.

The third is efficacy-failure category 3. Examples include malabsorptive bariatric surgery with oral contraception, rifampin with progestin-only pills, and enzyme-inducing antiepileptics with CHC or progestin-only pills.

These types should not be taught the same way.

In toxicity category 3, the method may add a harmful physiologic effect. In efficacy category 3, the method may fail. In risk-benefit category 3, the same method may offer a real benefit and a real danger at the same time.

That distinction changes counseling.

For moderate hypertension and CHC, the counseling is about vascular risk and choosing a non-estrogen method. For malabsorptive bariatric surgery and oral contraception, the counseling is about unreliable absorption and choosing a nonoral method. For anticoagulation and CHC, the counseling is about whether bleeding control benefit justifies estrogen in a therapeutically anticoagulated patient, often with LNG-IUD or DMPA as better options.

The bedside rule is this: category 3 is a stop-and-reason category. It does not end the conversation, but it prevents autopilot prescribing.

The Category 4 Mindset: The Mechanisms Converge Too Strongly

Category 4 is where the method and disease push in the same dangerous direction with unacceptable intensity.

The first 21 postpartum days plus CHC: postpartum hypercoagulability, stasis, tissue injury, and estrogen procoagulant effect.

Major surgery with prolonged immobilization plus CHC: surgical injury, venous stasis, and estrogen procoagulant effect.

Known familial thrombophilia plus CHC: impaired anticoagulant control or increased thrombin substrate plus estrogen-driven coagulation activation.

SLE with positive antiphospholipid antibodies plus CHC: immunothrombosis plus estrogen-driven coagulation activation.

Migraine with aura plus estrogen: neurovascular stroke phenotype plus estrogen-associated arterial thrombotic pressure.

Severe hypertension or vascular disease plus CHC: high-pressure arterial injury or established vascular disease plus estrogen RAAS and procoagulant effects.

Current or recent breast cancer plus hormonal contraception: active or recently treated hormone-sensitive malignancy plus exogenous hormone signaling.

Active intrauterine GTD with elevated hCG or suspected intrauterine disease plus new IUD placement: vascular friable trophoblastic disease plus instrumentation and foreign body.

Immediate postpartum uterine infection plus IUD insertion: infected reproductive tract and vulnerable postpartum cavity plus instrumentation and foreign body.

Immediate postpartum ongoing hemorrhage plus IUD insertion: unstable bleeding uterus plus nonessential intrauterine procedure.

Category 4 is not a punishment category. It is a convergence category. The method adds exactly the wrong stress to exactly the wrong substrate.

How To Reason Through A Patient At The Bedside

Now imagine the resident in clinic. The patient does not arrive as a table row. She arrives as a story.

Start by naming the method mechanism.

If the patient wants combined hormonal contraception, ask: is there a thrombotic, arterial, hypertensive, postpartum, migraine-with-aura, thrombophilia, antiphospholipid, or breast cancer substrate? If yes, estrogen may be the wrong signal.

If the patient wants DMPA, ask: is the issue arterial disease, severe hypertension, advanced diabetes, hypoestrogenic bone or lipid concern, or severe thrombocytopenia at initiation? If yes, DMPA's distinct profile matters.

If the patient wants a progestin-only pill, ask: will a low-dose oral method remain reliable? Rifampin, rifabutin, enzyme-inducing antiepileptics, malabsorption, vomiting, diarrhea, or adherence problems can move the concern from safety to failure.

If the patient wants the implant, ask: is there enzyme induction? The implant has more reserve than the pill but is not fully immune.

If the patient wants an LNG-IUD, ask two different questions. First, is there hormone-sensitive breast cancer? Second, is the uterine cavity safe for insertion: no active infection, no ongoing hemorrhage, no suspected active intrauterine GTD?

Then name the patient substrate.

Postpartum? Locate the day. First 21 days. Day 21 to 29. Day 30 to 42. More than 42 days. Breastfeeding or not. Other VTE risk factors or not.

Vascular disease? Separate venous from arterial. Prior VTE, thrombophilia, prior MI, prior stroke, hypertension, diabetes, migraine aura, obesity, smoking, age, vascular disease.

Immune disease? In SLE, always ask about antiphospholipid antibodies and severe thrombocytopenia. In immunosuppression, think infection risk with devices but do not invent risk when data are reassuring.

Cancer? Breast cancer is hormone signaling. Gynecologic cancer has a different logic. GTD is hCG surveillance and uterine cavity disease.

Medication interaction? Ask whether the drug lowers steroid exposure. Rifampin and enzyme-inducing antiepileptics matter. Broad-spectrum antibiotics, antifungals, and antiparasitics other than rifampin or rifabutin generally do not interfere with OC efficacy according to ACOG.

Finally, compare against pregnancy risk. In many medically complex patients, pregnancy is not a neutral alternative. Pregnancy may increase VTE risk, worsen diabetes risk, complicate anticoagulation, expose a fetus to warfarin or antiepileptic drugs, or disrupt hCG surveillance after GTD. A category 2 method may be safer than pregnancy. A category 3 method may still be considered if all safer options are unacceptable. But a category 4 pairing requires a very strong reason and usually a different method.

Case Walkthroughs To Seal The Model

A breastfeeding patient comes at 18 days postpartum asking for the pill she used before pregnancy, a combined pill. She has no thrombophilia, no smoking, and no prior VTE. The answer is still no. First 21 days postpartum is category 4 for CHC regardless of breastfeeding status. The reason is postpartum hypercoagulability, stasis, tissue injury, and estrogen hepatic coagulation effects. Offer a progestin-only method or IUD if no infection or hemorrhage contraindication exists, with counseling about postpartum IUD expulsion if relevant.

The same patient returns at 35 days postpartum. She is breastfeeding and has no other VTE risk factors. Now CHC is category 2. The first month has passed, lactation is more established, and no extra thrombotic driver is present. But it is not category 1 because postpartum VTE risk and breastfeeding physiology still matter.

Change one fact: she is 35 days postpartum, breastfeeding, and had a cesarean with BMI 35 and preeclampsia. Now CHC is category 3. The category changes because thrombosis is cumulative. Cesarean adds surgical injury, BMI adds stasis and inflammation, preeclampsia adds endothelial dysfunction, postpartum physiology remains active, and estrogen adds hepatic procoagulant drive.

A 27-year-old has migraine with visual aura and wants the patch. The patch is estrogen-containing. This is category 4. The issue is not headache pain. It is aura as a neurovascular stroke-risk phenotype plus estrogen-associated arterial thrombotic pressure.

A 27-year-old has menstrual migraine without aura, no smoking, no hypertension, no other stroke risk factors, and wants extended-cycle CHC. This is category 2. Migraine remains a neurovascular diagnosis, so not category 1. But the aura phenotype is absent, and hormonal stabilization may reduce estrogen-withdrawal migraine.

A 32-year-old has BP 148 over 94 on repeated measurements and wants CHC. This is category 3. Estrogen can increase hepatic angiotensinogen, support angiotensin II

and aldosterone effects, raise vascular resistance and sodium retention, impair endothelial function, and add procoagulant tone. At this BP range, she is not yet in category 4, but CHC should not be the default.

A 32-year-old has BP 164 over 104 and wants CHC. This is category 4. High-pressure arterial injury plus estrogen RAAS and coagulation effects creates unacceptable risk.

A 32-year-old has BP 164 over 104 and asks about DMPA. This is category 3, not 4. DMPA avoids estrogen's RAAS effect, but possible lipid and hypoestrogenic vascular effects matter in severe hypertension. Prefer safer alternatives if acceptable.

A patient with known factor V Leiden asks for CHC. Category 4. The anticoagulant brake is impaired, and estrogen increases coagulation drive. The interaction is not theoretical enough to ignore.

The same patient asks for LNG-IUD. Category 2. The thrombophilia remains real, so not category 1, but the LNG-IUD avoids estrogen and works mostly locally. Pregnancy would be hypercoagulable. The balance generally favors the method.

A patient with SLE asks for contraception. Do not answer yet. Ask about antiphospholipid antibodies, cardiovascular risk factors, and thrombocytopenia. If antiphospholipid antibodies are positive, CHC is category 4, and progestin-only methods including LNG-IUD are category 3. If antibodies are negative, no cardiovascular risk factors are present, and disease is stable, CHC may be category 2. If she has severe thrombocytopenia under 50,000 and asks to start DMPA, initiation is category 3 because early irregular bleeding can be clinically important when platelet-mediated hemostasis is impaired.

A patient treated for breast cancer last year asks for LNG-IUD. Current or recent breast cancer is category 4 for hormonal contraception, including LNG-IUD. The issue is not uterine safety. It is systemic oncologic caution in a hormone-sensitive disease context. Copper IUD is the category 1 contrast.

A patient treated for breast cancer eight years ago with no evidence of disease asks for hormonal contraception. Category 3. The disease-free interval lowers concern compared with current or recent disease, but residual hormone-sensitive recurrence risk is not zero. Nonhormonal contraception is preferred unless individualized oncology-informed reasoning supports hormonal use.

A patient after Roux-en-Y gastric bypass asks for oral contraception. Category 3. The problem is not hormone toxicity. The problem is unreliable absorption. Use a nonoral method.

A patient taking rifampin asks for progestin-only pills. Category 3. Rifampin can induce steroid metabolism enough that a low-dose oral progestin method may fail. DMPA or LNG-IUD is category 1 and should be encouraged for long-term rifamycin use.

A patient taking carbamazepine asks for the implant. Category 2. The interaction is real: etonogestrel levels can fall and pregnancies have been reported. But the implant has high baseline efficacy and continuous release. If maximal reliability is needed, DMPA or IUD is mechanistically stronger.

A patient on therapeutic anticoagulation with heavy menstrual bleeding asks for CHC because it controlled her bleeding in the past. This is category 3, but not a thoughtless no. CHC can reduce bleeding and suppress ovulation, which may help hemorrhagic ovarian cyst risk. But it adds estrogenic thrombosis biology to a patient anticoagulated for a thrombotic indication. The safer reasoning usually starts with LNG-IUD or DMPA, but CHC can be considered case by case if alternatives are unacceptable and anticoagulation is reliable.

Final Synthesis

The durable model is this.

Category 2 means there is a real issue, but it is bounded. The method avoids the main dangerous mechanism, the exposure is local or buffered, the timing is safer, or the competing risk of pregnancy is larger.

Category 3 means the risk usually wins. Sometimes the method may harm the patient physiologically. Sometimes the method may fail. Sometimes the method offers a real benefit and a real danger at the same time. Category 3 demands individualized reasoning and usually a different default method.

Category 4 means the method adds the wrong mechanism to an already dangerous substrate. Estrogen plus peak postpartum clotting. Estrogen plus thrombophilia. Estrogen plus antiphospholipid immunothrombosis. Estrogen plus migraine aura. Estrogen plus severe hypertension. Hormones plus current or recent breast cancer. IUD insertion into infection, hemorrhage, or active intrauterine GTD.

Do not memorize category numbers as isolated facts. Build the category from the mechanism.

Name the substrate. Name the method effect. Ask whether they converge. Ask whether the risk is bounded, usually unfavorable, or unacceptable. Then choose the safest method that still respects the patient's goals.

That is how ACOG 206 becomes usable at the bedside.

Official Sources

This transcript is based on ACOG Practice Bulletin No. 206, *Use of Hormonal Contraception in Women With Coexisting Medical Conditions*, and the U.S. Medical Eligibility Criteria framework for contraceptive use.

Official sources referenced:

- ACOG Practice Bulletin No. 206: *Use of Hormonal Contraception in Women With Coexisting Medical Conditions*.
- U.S. Medical Eligibility Criteria for Contraceptive Use.